Three copies of the Werner state: the sharp curve in dimensions three and four and a bound in every dimension

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Let $\rho_\alpha = (I - \alpha F)/d(d-\alpha)$ be the Werner state on $\mathbb{C}^d \otimes \mathbb{C}^d$. It is NPT for $\alpha > 1/d$ and one-copy distillable for $\alpha > 1/2$; whether the states in between are distillable from any number of copies is the NPT bound-entanglement problem. Exact rational certificates published in 2026 show that for $d = 3, 4$ three copies do not suffice at $\alpha = 1/2$: the form $q(C) = \sum_S (-\frac{1}{2})^{|S|} \| \operatorname{Tr}_S C \|^2$ is nonnegative on every $d^3 \times d^3$ matrix of rank at most two. From this single computer-verified input we prove by hand that for $0 \le \alpha \le 1/2$ the minimum of the corresponding form on unit matrices of rank at most two is exactly $(1-2\alpha)(1-\alpha)^2$, determine all minimisers for $0 < \alpha < 1/2$, exhibit three families of endpoint zeros and conjecture that they are all, and derive operational and higher-dimensional consequences. A second, dimension-uniform certificate—built from singular frames and the walled Brauer algebra, with a quantitative endpoint bound proved by hand as its only hand input—shows that three copies do not suffice for $0 \le \alpha \le \alpha_0 = 0.4625$ in every finite local dimension, so NPT Werner states that survive three copies exist for every $d \ge 3$. We explain both certificates and state exactly what the computer checks.

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Results were formalized in Lean 4 by AI agents and checked by Lean's kernel; the formalization is at https://github.com/physical-superintelligence/oqp-batch-2026/tree/main/oqp02-three-copy-werner/lean.

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I. INTRODUCTION

A bipartite state is *distillable* if local operations and classical communication can extract maximally entangled qubit pairs from many copies of it. States with positive partial transpose (PPT) are not distillable [3]; whether every state with negative partial transpose (NPT) is distillable has been open since 1999, when [4, 5] gave evidence that it fails for the Werner states [6] (see [7] for a review). With F the swap on $\mathbb{C}^d \otimes \mathbb{C}^d$,

$$\rho_\alpha = \frac{I - \alpha F}{d(d-\alpha)}, \qquad -1 \le \alpha \le 1, \tag{1}$$

is NPT exactly when $\alpha > 1/d$ and one-copy distillable exactly when $\alpha > 1/2$. The window $1/d < \alpha \le 1/2$, nonempty for $d \ge 3$, is the arena of the conjecture that no finite number of copies distils these states; since every NPT state can be brought to an NPT Werner state by local filtering and twirling [8], the Werner family is the whole problem (Problem 4 of [9]). Beyond one copy only partial results [4, 5, 10–13] were known until July 2026, when two copies were settled in all dimensions by four

* I.M. chose the three-copy Werner problem as the arena for the NPT bound-entanglement question, scoped the project, and designed, ran and steered the automated research workflows at PSI over several months, including the campaigns in which the parameter $\alpha_0 \approx 0.4627$ of Theorem 1.3 was first reached (the certificate presented here is a fresh derivation of that result). I.M. set the methodological directions that shaped the paper: that the hand mathematics should carry as much of the argument as possible and the computer as little; that certificates must couple the three sites through one lifted object rather than treat sectors separately; that the published endpoint certificates be used as inputs and replayed rather than reproduced; and that every computer-verified statement be stated as such with a minimal, re-runnable package. I.M. directed the editorial process (two independent drafts, an adversarial editorial debate between AI referees, and two rounds of referee revision), obtained and supplied a referee report on an earlier version, decided which of its recommendations to adopt, and reviewed and edited the final text.; ian@psi.inc

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independent groups [14–17]. For three copies, Wu and Zou [18] proved the case of normal test matrices, and exact computer certificates for the general case appeared in October 2026 for $d = 3$ [1] and $d = 4$ [2].

The n-copy question is a question about matrices. For C on $\mathcal{H} = (\mathbb{C}^d)^{\otimes 3}$ let $\operatorname{Tr}_S C$ be its partial trace over the sites in $S \subseteq \{1,2,3\}$ and

$$q_\alpha(C) = \sum_{S\subseteq\{1,2,3\}} (-\alpha)^{|S|} \,\|\operatorname{Tr}_S C\|^2, \qquad q := q_{1/2}. \quad (2)$$

Then ρ_α is three-copy distillable if and only if $q_\alpha(C) < 0$ for some C of rank at most two (Proposition 2.1). The published certificates establish:

Theorem 1.1 (endpoint; [1] for $d = 3$, [2] for $d = 4$)**.** *Let $d \in \{3, 4\}$. Then $q(C) \geq 0$ for every $C \in \operatorname{End}(\mathcal{H})$ of rank at most two.*

By Lemma 2.4(b) the theorem holds for unequal local dimensions $2 \leq d_1, d_2, d_3 \leq 4$, and the case $d = 3$ is also a corollary of the case $d = 4$, so the qutrit endpoint is established twice (Remark 6.5). The endpoint inequality is the result of [1, 2] and is used as such; the threshold $\alpha \leq \frac{1}{2}$ already follows from it by a convex decomposition (Remark 3.2). New here are the exact constant, the interior minimisers, the zero families (which extend the endpoint examples of [18]) and the consequences:

Theorem 1.2 (the three-copy curve)**.** *For local dimensions $2 \leq d_i \leq 4$ and $0 \leq \alpha \leq \frac{1}{2}$,*

$$\min\{q_\alpha(C) : \ \|C\| = 1, \ \operatorname{rank} C \leq 2\} = (1-2\alpha)(1-\alpha)^2, \quad (3)$$

and for $0 < \alpha < \frac{1}{2}$ the minimisers are exactly $2^{-1/2} P_E \otimes |u\rangle\langle u| \otimes |v\rangle\langle v|$ up to a phase and a permutation of the sites, with P_E the projector onto a two-plane E of one local space and u, v unit vectors in the other two. Consequently ρ_α is three-copy distillable if and only if $\alpha > \frac{1}{2}$.

The mechanism is an exact identity (Proposition 3.1) that writes $q_\alpha - (1-2\alpha)(1-\alpha)^2\|C\|^2$ as a positive combination of the endpoint form q, the one-site form and the two-site forms on the three cuts of C; the single endpoint inequality therefore determines the whole curve, and the equality analysis gives the minimisers. An operational form (every two-qubit state obtained from three copies by a separable operation is separable), an extension to arbitrary dimensions when the local overlaps of range and corange have rank at most four, full-rank NPT states that survive three copies, and a quantitative robustness statement that uses the constant (Sec. V) follow. At the endpoint the equality set is larger than the minimisers; we exhibit three families of zeros and conjecture, with evidence, that they exhaust it (Sec. IV).

In dimensions $d \geq 5$ the endpoint is open. A second certificate, uniform in the dimension, gives the sign on a slightly shorter interval:

Theorem 1.3 (every dimension)**.** *Let $\alpha_0 = \frac{37}{80}$. For all finite local dimensions d_1, d_2, d_3 and $0 \leq \alpha \leq \alpha_0$,*

$$q_\alpha(C) \geq (1 - \alpha/\alpha_0)^3 \,\|C\|^2 \quad (4)$$

for every C of rank at most two. Consequently $\rho_\alpha^{(d)}$ is three-copy undistillable for $0 \leq \alpha \leq \alpha_0$ in every dimension d, and the NPT window $1/d < \alpha \leq \alpha_0$ of states that survive three copies is nonempty for every $d \geq 3$.

The certificate behind Theorem 1.3 (Sec. VII) lifts C by its singular frames, replaces the moment matrix of Sec. VI by a 2×2 determinant, and tests positivity in the walled Brauer algebra, whose structure constants are polynomials in d; its only hand input is a quantitative bound on the endpoint form valid in every dimension, proved by hand in Appendix B, and the value of α_0 is what four replicas give in this way, not an optimality claim. The gap $\alpha_0 < \alpha \leq \frac{1}{2}$ for $d \geq 5$ remains (Sec. IX).

Section VI explains the endpoint certificates: it derives by hand the lift of C to a pair of vectors on which "rank at most two" is a linear condition, the expression of q as a swap polynomial, and the averaging that makes the problem finite, so that Theorem 1.1 becomes a four-line deduction from an identity between explicitly stored rational matrices; Sec. VII does the same for the uniform certificate, and Sec. VIII states exactly which finite statements the computer checks. Every other numbered statement is proved in the paper, except Theorem 2.3 (cited), Theorems 6.4 and 7.2 (computer-verified, Sec. VIII) and Conjecture 4.4. Section II fixes conventions and derives the matrix form, Sec. III proves Theorem 1.2, Sec. IX states what remains open, and Appendix A writes the two-site theorem for rank two as an explicit sum of squares.

II. DISTILLABILITY AS A MATRIX INEQUALITY

Conventions. Inner products are conjugate-linear in the first argument, $\|\cdot\|$ is the Frobenius norm and $\|\cdot\|_1$ the trace norm. $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$ with $\mathcal{H}_i = \mathbb{C}^{d_i}$; the factors are the *sites*, and $d_i = d$ unless said otherwise. $\operatorname{Tr}_S C$ is the unnormalised partial trace over the sites in S ($\operatorname{Tr}_\emptyset C = C$). The expression (2) on a tensor product of n factors defines the n-site form $q_\alpha^{(n)}$, with $q^{(n)} := q_{1/2}^{(n)}$; thus $q_\alpha^{(1)}(C) = \|C\|^2 - \alpha|\operatorname{Tr} C|^2$ on one site, and on a bipartition $\mathcal{H}_L \otimes \mathcal{H}_R$ we write

$$Q_{L|R}(C) = \|C\|^2 - \tfrac{1}{2}\|\operatorname{Tr}_L C\|^2 - \tfrac{1}{2}\|\operatorname{Tr}_R C\|^2 + \tfrac{1}{4}|\operatorname{Tr} C|^2 \quad (5)$$

for the two-site form $q^{(2)}$. "Rank" means the rank of C as a $d^3 \times d^3$ matrix; a matrix of rank at most two is $C = XY^\dagger$ with $X, Y \in \mathbb{C}^{d^3\times 2}$, and nothing else is assumed (C need not be normal, its singular values may differ, its range and corange may be different planes). Row-first vectorisation sends $C = \sum_{ab} C_{ab}|a\rangle\langle b|$ to $\operatorname{vec} C =$

$\sum_{ab} C_{ab}|a\rangle|b\rangle \in \mathcal{H}\otimes\mathcal{H}$, whose Schmidt rank across the two copies of $\mathcal{H}$ is $\operatorname{rank} C$. On $\mathbb{C}^d\otimes\mathbb{C}^d$ let $|\Omega\rangle = \sum_j |jj\rangle$, $\Omega = |\Omega\rangle\langle\Omega|$, F the swap and Γ the partial transpose of the second factor, so $F^\Gamma = \Omega$.

Werner states and distillability. We write $\rho_\alpha^{(d)}$ for (1) when the dimension matters; the parameter of [1, 2] is $-\alpha$. Since $\rho_\alpha^\Gamma = (I-\alpha\Omega)/d(d-\alpha)$ and Ω has the eigenvalue d on $|\Omega\rangle$, ρ_α is NPT exactly when $\alpha > 1/d$, and it is separable exactly when it is PPT [6, 19]. A state ϱ is *one-copy distillable* if $\langle\psi, \varrho^\Gamma\psi\rangle < 0$ for some vector ψ of Schmidt rank at most two across the Alice–Bob cut; equivalently [3–5], local projections onto a two-qubit subspace turn one copy, with nonzero probability, into an entangled two-qubit state. It is *n-copy distillable* if $\varrho^{\otimes n}$ is one-copy distillable, and *distillable* if it is n-copy distillable for some n: a distillable state yields maximally entangled qubit pairs asymptotically from many such blocks, and conversely [3]. PPT states are not distillable.

Proposition 2.1 (matrix form)**.** *For every $C \in \operatorname{End}(\mathcal{H})$ and real α,*

$$\Big\langle \operatorname{vec} C,\ \bigotimes\nolimits_i \big(I - \alpha\Omega_i\big)\ \operatorname{vec} C \Big\rangle = q_\alpha(C), \tag{6}$$

where Ω_i acts on the site-i factors of the two copies of $\mathcal{H}$. For $-1 \le \alpha \le 1$, $(\rho_\alpha^{(d)})^\Gamma = (I-\alpha\Omega)/d(d-\alpha)$, so the left side with $(\rho_\alpha^{(d_i)})^\Gamma$ in place of $I-\alpha\Omega_i$ equals $q_\alpha(C)/\prod_i d_i(d_i-\alpha)$, and $\bigotimes_i \rho_\alpha^{(d_i)}$, in particular $\rho_\alpha^{\otimes 3}$, is one-copy distillable if and only if $q_\alpha(C) < 0$ for some C of rank at most two.

Proof. Pair the site-i factor of the first copy of $\mathcal{H}$ (the row index of C) with the site-i factor of the second copy (the column index). Contracting $\operatorname{vec} C$ with $\langle\Omega|$ on such a pair sets the row index equal to the column index at site i and sums, i.e. takes the partial trace over site i; hence $\langle \operatorname{vec} C, \Omega_S \operatorname{vec} C\rangle = \|\operatorname{Tr}_S C\|^2$ for $\Omega_S = \prod_{i\in S}\Omega_i$, and expanding the product gives (6). The second statement is the criterion above, since the Schmidt rank of $\operatorname{vec} C$ is $\operatorname{rank} C$. □

The three-copy problem is therefore a statement about $d^3\times d^3$ matrices of rank at most two. Its difficulty is visible in (2): partial traces do not preserve rank ($\operatorname{Tr}_1 C$ can have rank $2d$), so lower-copy results cannot simply be applied to partial traces; a proof has to use that all columns of C lie in one plane and all rows in one plane, at all three sites at once.

Lemma 2.2 (one site)**.** *If $\operatorname{rank} C \le 2$ then $|\operatorname{Tr} C| \le \|C\|_1 \le \sqrt{2}\,\|C\|$; hence $q_\alpha^{(1)}(C) \ge (1-2\alpha)\|C\|^2$ for $\alpha \ge 0$, with equality for $\alpha > 0$ if and only if C is a complex multiple of an orthogonal projector of rank two. In particular, by the one-site case of Proposition 2.1, ρ_α is one-copy distillable iff $\alpha > \frac12$.*

Proof. With singular values $s_1 \ge s_2$, $|\operatorname{Tr} C| \le s_1 + s_2 = \|C\|_1$ and $(s_1+s_2)^2 \le 2(s_1^2+s_2^2)$. Equality in the second inequality means $s_1 = s_2$; in the first, with the polar decomposition $C = U|C|$, it means $|\operatorname{Tr}(U|C|)| = \operatorname{Tr}|C|$, i.e. U is a constant phase on $\operatorname{ran}|C|$. □

Theorem 2.3 (two sites [14–17])**.** *$Q_{L|R}(C) \ge 0$ for every C of rank at most two on $\mathcal{H}_L\otimes\mathcal{H}_R$, in all dimensions.*

The four proofs are independent; [16, 17] also give sharp forms. Appendix A writes the inequality for rank two as a sum of squares.

Lemma 2.4. (a) *If C_1 acts on a set of sites and C_2 on the complementary sites, then $q_\alpha^{(n)}(C_1\otimes C_2) = q_\alpha^{(n_1)}(C_1)\, q_\alpha^{(n_2)}(C_2)$.* (b) *If $V = \bigotimes_i V_i$ with isometries $V_i : \mathbb{C}^{k_i} \to \mathbb{C}^{d_i}$, then $q_\alpha(VCV^\dagger) = q_\alpha(C)$ and $\operatorname{rank} VCV^\dagger = \operatorname{rank} C$.*

Proof. (a) If C_1 acts on the sites in S_1, then $\operatorname{Tr}_S(C_1\otimes C_2) = \operatorname{Tr}_{S\cap S_1} C_1 \otimes \operatorname{Tr}_{S\setminus S_1} C_2$, and Frobenius norms multiply. (b) $\operatorname{Tr}_S(VCV^\dagger) = V'(\operatorname{Tr}_S C)V'^\dagger$ with V' the tensor product of the V_i over the sites not in S, and isometries preserve Frobenius norms. □

By (b), nothing depends on the choice of local bases.

III. FROM THE ENDPOINT TO THE WHOLE INTERVAL

Let $\bar Q(C) = \frac13\sum_{i=1}^3 Q_{i|\bar i}(C)$ be the mean of the two-site forms (5) over the three cuts $i\,|\,\bar i$ of $\mathcal{H}$, $\bar i = \{1,2,3\}\setminus\{i\}$.

Proposition 3.1 (continuation identity)**.** *For every $C \in \operatorname{End}(\mathcal{H})$ and every real α,*

$$q_\alpha(C) = (1-2\alpha)(1-\alpha)^2\|C\|^2 + \frac{4\alpha(1+\alpha)}{3}\, q(C) + \alpha(1-2\alpha)\Big(\frac23 - \alpha\Big) q^{(1)}(C) + 2\alpha(1-2\alpha)\,\bar Q(C), \tag{7}$$

where $q^{(1)}(C) = \|C\|^2 - \frac12|\operatorname{Tr} C|^2$ is the one-site form of C regarded as a matrix on the single site $\mathcal{H}$.

Proof. Write $t_k = \sum_{|S|=k}\|\operatorname{Tr}_S C\|^2$. Then $q_\alpha = \sum_k(-\alpha)^k t_k$, $q = t_0 - \frac12 t_1 + \frac14 t_2 - \frac18 t_3$, $q^{(1)} = t_0 - \frac12 t_3$, and $\bar Q = t_0 - \frac16(t_1+t_2) + \frac14 t_3$, because each single-site and each two-site trace occurs in exactly one of the three cuts. Comparing the coefficients of t_1, t_2, t_3 on the two sides of (7) gives three linear equations for the three multipliers, with the displayed solution; the coefficient of t_0 is then $1 - \frac43\alpha(1+\alpha) - \alpha(1-2\alpha)(\frac23-\alpha) - 2\alpha(1-2\alpha) = (1-2\alpha)(1-\alpha)^2$. □

On $[0, \frac12]$ the three multipliers in (7) are nonnegative, and on $(0, \frac12)$ strictly positive. The three forms they multiply are nonnegative on matrices of rank at most two: q by Theorem 1.1, $q^{(1)}$ by Lemma 2.2, $\bar Q$ by Theorem 2.3.

Proof of Theorem 1.2. Lower bound. Immediate from (7) and the three nonnegativity statements.

Sharpness. The matrix $2^{-1/2}P_E \otimes |u\rangle\langle u| \otimes |v\rangle\langle v|$ has norm one and rank two, and by Lemma 2.4(a) and the one-site values $q_\alpha^{(1)}(2^{-1/2}P_E) = 1-2\alpha$, $q_\alpha^{(1)}(|u\rangle\langle u|) = 1-\alpha$, its q_α equals $(1-2\alpha)(1-\alpha)^2$.

Minimisers. Let $0 < \alpha < \frac{1}{2}$ and $q_\alpha(C) = (1-2\alpha)(1-\alpha)^2$ with $\|C\| = 1$. All three multipliers in (7) are positive, so $q^{(1)}(C) = 0$ and $\bar{Q}(C) = 0$. By Lemma 2.2, $C = 2^{-1/2}P$ up to a phase, with P an orthogonal projector of rank two. Since each $Q_{i|\bar{i}}(P) \geq 0$, $\bar{Q}(P) = 0$ forces $Q_{i|\bar{i}}(P) = 0$ for $i = 1,2,3$. Let u_1, u_2 be an orthonormal basis of $\operatorname{ran} P$, and on two copies of $\mathcal{H}$ let F_i exchange the site-i factors. The swap trick $\|\operatorname{Tr}_{\bar{i}} P\|^2 = \operatorname{Tr}[(P\otimes P)F_i]$ and $\operatorname{Tr}[(P\otimes P)F_iF_{\bar{i}}] = \operatorname{Tr} P^2 = 2$ give

$$\begin{aligned} Q_{i|\bar{i}}(P) &= 3 - \tfrac{1}{2}\operatorname{Tr}\big[(P\otimes P)(F_i + F_{\bar{i}})\big] \\ &= \tfrac{1}{2}\operatorname{Tr}\big[(P\otimes P)(I-F_i)(I-F_{\bar{i}})\big] \\ &= 2\sum_{r,s=1}^{2} \|J(u_r\otimes u_s)\|^2, \end{aligned} \tag{8}$$

where $J = \frac{1}{4}(I-F_i)(I-F_{\bar{i}})$ is an orthogonal projector (F_i and $F_{\bar{i}}$ commute). Hence $J(w\otimes w) = 0$ for every $w \in \operatorname{ran} P$. For a unit vector w, expanding $4J$ into its four swap words gives $\|J(w\otimes w)\|^2 = \frac{1}{2}(1 - \text{purity of the site-}i\text{ marginal of } w)$, which vanishes iff w is a product vector across the cut $i|\bar{i}$. So every $w \in \operatorname{ran} P$ is a product at every cut, i.e. fully product. Two independent fully product vectors whose factors differ at more than one site have a sum of Schmidt rank two across one of the cuts; hence the spanning vectors of $\operatorname{ran} P$ differ at only one site, $\operatorname{ran} P = E\otimes u\otimes v$ up to a permutation of the sites, and C is of the stated form.

Distillability. For $\alpha \leq 0$ the state is PPT; for $0 < \alpha \leq \frac{1}{2}$, $q_\alpha \geq 0$ on rank two and Proposition 2.1 applies; for $\alpha > \frac{1}{2}$ the state is already one-copy distillable (Lemma 2.2), and tensoring a one-site witness with $|0\rangle\langle 1|\otimes|0\rangle\langle 1|$ gives a three-site witness by Lemma 2.4(a). □

Remark 3.2. (a) The threshold alone needs no identity: $\rho_\alpha = \lambda\rho_{1/2} + (1-\lambda)I/d^2$ with $\lambda = \alpha(2d-1)/(d-\alpha) \in [0,1]$ for $0 \leq \alpha \leq \frac{1}{2}$. Since I/d^2 is a nonnegative combination of product projectors $|j\rangle\langle j|\otimes|k\rangle\langle k|$, expanding $(\rho_\alpha^{\otimes 3})^\Gamma$ and contracting $\operatorname{vec} C$ with such a projector at a site compresses C by a row and a column vector there, which preserves rank at most two; what remains is an endpoint form on the other sites, nonnegative by Theorem 1.1, Theorem 2.3 or Lemma 2.2. This gives only $q_\alpha \geq (1-2\alpha)^3\|C\|^2$; the identity (7) gives the exact constant $(1-2\alpha)(1-\alpha)^2$ and the minimisers. (b) The proof uses only Lemma 2.2, Theorem 2.3 and the endpoint inequality, so Theorem 1.2 holds in any local dimensions in which the endpoint inequality holds, provided some $d_i \geq 2$ (the minimiser needs one site of dimension at least two; the lower bound itself holds in all dimensions). (c) At $\alpha = \frac{1}{2}$ the minimisers remain zeros of q, but they are no longer the only ones: $q(2^{-1/2}P_E\otimes N) = 0$ for every rank-one N on the other two sites, and there are zeros with two-dimensional local support at every site (Sec. IV).

IV. ENDPOINT ZEROS

Every nonzero zero of q of rank at most two has rank two: for $C = xy^\dagger$, $\|\operatorname{Tr}_S C\|^2 = \langle x\otimes y, \prod_{i\in S} F_i\, x\otimes y\rangle$ with F_i as in Sec. III, so $q(C) = \langle x\otimes y, \prod_i (I - \frac{1}{2}F_i)\, x\otimes y\rangle \geq \frac{1}{8}\|x\|^2\|y\|^2$, the three commuting swaps having eigenvalues ± 1 [13].

Let θ be the spin flip on r qubits, $\theta\psi = \Xi\bar{\psi}$ with $\Xi = (i\sigma^y)^{\otimes r}$, an antiunitary with $\theta^2 = (-1)^r$. For orthonormal $x, y \in (\mathbb{C}^2)^{\otimes r}$ put

$$\beta(x,y) = \tfrac{1}{\sqrt{2}}\big(|x\rangle\langle\theta y| - |y\rangle\langle\theta x|\big) = \tfrac{1}{\sqrt{2}}\,(xy^{\mathsf{T}} - yx^{\mathsf{T}})\,\Xi^{\mathsf{T}}, \tag{9}$$

the Plücker bivector of the plane $\operatorname{span}\{x,y\}$, viewed as an antisymmetric matrix, composed with the local spin flips. Replacing (x,y) by another frame of the same plane multiplies β by a nonzero scalar. A *binary embedding* is $V = V_1\otimes V_2\otimes V_3$ with isometries $V_i : \mathbb{C}^2 \to \mathbb{C}^d$.

Definition 4.1 (three families)**.** Up to a phase and a permutation of the sites:

$$\text{I: } 2^{-1/2}P_E\otimes|\varphi\rangle\langle\chi|, \qquad \text{II: } |u\rangle\langle v|\otimes V\beta(x,y)V^\dagger, \qquad \text{III: } V\beta(x,y)V^\dagger,$$

where $E \subset \mathbb{C}^d$ is a two-plane, $\varphi, \chi \in \mathbb{C}^d\otimes\mathbb{C}^d$ and $u, v \in \mathbb{C}^d$ are unit vectors, $x, y \in (\mathbb{C}^2)^{\otimes 2}$ in family II and $(\mathbb{C}^2)^{\otimes 3}$ in family III, and $V = V_2\otimes V_3$ in family II.

Up to normalisation, family I contains the non-normal equality operator $|0\rangle\langle 1|\otimes|0\rangle\langle 1|\otimes I_2$ of [18, Prop. 8.1], and family III contains the balanced anti-state mixtures $|x\rangle\langle x| + |\theta x\rangle\langle\theta x|$ of [18, Prop. 8.2]—among them the projector onto the span of $|000\rangle$ and $|111\rangle$—since $\beta(x,\theta x) = -2^{-1/2}(|x\rangle\langle x| + |\theta x\rangle\langle\theta x|)$ for odd r. The extension to arbitrary two-planes is new; it produces non-normal members such as the nilpotent $\sigma_2^z(\sigma_1^z + \sigma_3^z) + i(\sigma_1^x\sigma_3^x - \sigma_1^y\sigma_3^y)$ on three qubits.

Proposition 4.2. (a) *Every member of the families I–III is a zero of q of rank two with both singular values equal to $2^{-1/2}$.* (b) *For $d = 3$, modulo the phase, families I and III have real dimension 36 and family II has dimension 24, and the union of the three families is path-connected (a stratified set, not a manifold).*

Proof. (a) Family I: by Lemma 2.4(a), $q(2^{-1/2}P_E\otimes|\varphi\rangle\langle\chi|)$ is the product of $q^{(1)}(2^{-1/2}P_E) = 1 - \frac{1}{2}\cdot 2 = 0$ with $Q_{2|3}(|\varphi\rangle\langle\chi|)$. Families II and III: on r qubits the one-site endpoint operator $I - \frac{1}{2}\Omega$ on $\mathbb{C}^2\otimes\mathbb{C}^2$ is positive semidefinite with kernel $\mathbb{C}|\Omega\rangle$ (its eigenvalue on $|\Omega\rangle$ is $1 - \frac{1}{2}\cdot 2 = 0$), so by Proposition 2.1 $q^{(r)}(N) = 0$ iff $\operatorname{vec} N \in \ker(I - \frac{1}{2}\Omega)^{\otimes r} = \sum_i |\Omega_i\rangle\otimes(\text{anything})$, i.e. iff

N has no component on a Pauli word of full weight r. Now $xy^{\mathsf T}-yx^{\mathsf T}$ is antisymmetric, hence a combination of Pauli words containing an odd number of factors σ^y (the only antisymmetric Pauli matrix), and right multiplication by $\Xi^{\mathsf T}=(-i\sigma^y)^{\otimes r}$ turns each σ^y into I and each other factor into a non-identity Pauli matrix. So every word of $\beta(x,y)$ contains an identity factor, and $\beta(x,y)$ is a zero of $q^{(r)}$. Moreover $\Xi^{\mathsf T}$ is unitary and $x\perp y$, so $\beta\beta^\dagger=\frac12(xy^{\mathsf T}-yx^{\mathsf T})(xy^{\mathsf T}-yx^{\mathsf T})^\dagger=\frac12(|x\rangle\langle x|+|y\rangle\langle y|)$: the singular values are $2^{-1/2},2^{-1/2}$. Lemma 2.4 transports these zeros to $\mathcal H$, the one-site factor $|u\rangle\langle v|$ of family II contributing the finite factor $q^{(1)}(|u\rangle\langle v|)$.

(b) The parameters of family I are a two-plane in $\mathbb C^3$ and two rays in $\mathbb C^9$ ($4+16+16=36$ real dimensions), of family III three local two-planes and a two-plane in $(\mathbb C^2)^{\otimes3}$ ($12+24=36$), of family II two rays in $\mathbb C^3$, two local two-planes and a two-plane in $(\mathbb C^2)^{\otimes2}$ ($8+8+8=24$); a generic member determines its parameters (the local two-planes are its local supports, and the remaining data are then determined), so these are the dimensions; non-generic members, such as the meeting points below, form lower-dimensional strata. Each parameter space is connected, family III is invariant under site permutations, and the families meet: writing β_r for β on r qubits, $\beta_3(|000\rangle,|100\rangle)=2^{-1/2}I_2\otimes|00\rangle\langle11|$ lies in family I, and $\beta_3(|0\rangle\otimes x',|0\rangle\otimes y')=-|0\rangle\langle1|\otimes\beta_2(x',y')$ with $x',y'\in(\mathbb C^2)^{\otimes2}$ lies in family II; so every placement of I and II is connected to III. □

Remark 4.3 (parity of the spin flip). For odd r, $\theta^2=-1$ gives $\langle\theta x,x\rangle=0$ and a short computation gives $\beta(x,y)^2=\frac12\big(\operatorname{Tr}\beta(x,y)\big)\beta(x,y)$: a family-III zero is a scaled oblique projector when $\operatorname{Tr}\beta\neq0$ and a square-zero nilpotent when $\operatorname{Tr}\beta=0$ (the example above), and it is non-normal in general. For even r, $\langle\theta x,y\rangle=\langle\theta y,x\rangle$ gives $\operatorname{Tr}\beta=0$, so β^2 is never a nonzero multiple of β; the simplest instance is $2^{-1/2}(|00\rangle\langle00|-|11\rangle\langle11|)$, whose square is $\frac12(|00\rangle\langle00|+|11\rangle\langle11|)$. The same construction gives rank-two zeros of the r-site endpoint form on r qubits for every r, of alternating algebraic type.

Conjecture 4.4. *For $d\in\{3,4\}$, every unit matrix of rank at most two with $q(C)=0$ belongs, up to a phase and a permutation of the sites, to one of the families I–III.*

Remark 4.5. (a) Random-start minimisation of $q/\|C\|^2$ over $(X,Y)\in\mathbb C^{d^3\times2}\times\mathbb C^{d^3\times2}$—i.i.d. complex Gaussian starts, L-BFGS with the analytic gradient (gradient tolerance 10^{-12}, at most 20 000 iterations), a run counted as a zero when $q/\|C\|^2<10^{-10}$—gave 74/6/420 zeros in families I/II/III for 500 starts at $d=3$ and 82/7/111 for 200 starts at $d=4$, with no unconverged run and no zero outside the families. Membership was decided by the local supports of range and corange at each site followed by a comparison with the normal forms of Definition 4.1 (relative residual below 10^{-2}; a matrix outside the families has residual of order one; script, seeds and logs in [20]). (b) At one sampled point of each family (seeds in [20]) the Hessian of $q/\|C\|^2$ in the parametrisation (X,Y) has nullity $46,34,46$ for I, II, III at $d=3$ and $78,46,58$ at $d=4$, with a clear spectral gap: in each case the dimension of the family ($36,24,36$ at $d=3$ by Proposition 4.2(b); $68,36,48$ at $d=4$ by the same count) plus the ten directions along which $q/\|C\|^2$ is constant, eight from $(X,Y)\mapsto(XR,YR^{-\dagger})$ with $R\in GL(2,\mathbb C)$ and two from the complex rescaling of C. This is an observation at sampled points: to second order nothing leaves the families there; it excludes neither a larger nullity at special points nor zeros elsewhere. (c) At a zero every term of the certificate identity (17) vanishes, so $G_t^{1/2}\widehat Y_t(\xi)G_t^{1/2}=0$, i.e. $\operatorname{ran}\widehat Y_t(\xi)\subseteq\ker G_t$ (for $d=4$, where the G_t are positive definite, every $\widehat Y_t(\xi)$ vanishes); a classification would require the joint kernel on the physical moment set, which the certificate does not provide.

V. CONSEQUENCES

Corollary 5.1 (operational form). *Let $d\in\{3,4\}$ and $\alpha\le\frac12$, or d arbitrary and $\alpha\le\alpha_0$. Every two-qubit state σ obtained from $\rho_\alpha^{\otimes3}$ by a separable operation (in particular by LOCC) with nonzero probability is separable; in particular its singlet fraction $\max_\Phi\langle\Phi|\sigma|\Phi\rangle$ over maximally entangled Φ is at most $\frac12$, the value of a product state.*

Proof. A separable operation has product Kraus operators $L_A^j\otimes L_B^j$, and the conditioned output is $\sigma=p^{-1}\sum_j(L_A^j\otimes L_B^j)\rho_\alpha^{\otimes3}(L_A^j\otimes L_B^j)^\dagger$ with $p>0$. Pairing σ^Γ with a two-qubit vector ψ gives $p^{-1}\sum_j\langle\psi_j,(\rho_\alpha^{\otimes3})^\Gamma\psi_j\rangle$ with $\psi_j=(L_A^{j\dagger}\otimes L_B^{j\mathsf T})\psi$ of Schmidt rank at most two, which is nonnegative by Remark 3.2(a) for $0\le\alpha\le\frac12$ (by Theorem 1.3 for $0\le\alpha\le\alpha_0$ in general dimension) and because ρ_α is PPT for $\alpha\le0$. So σ is PPT, hence separable [19, 21], and separable two-qubit states have singlet fraction at most $\frac12$ [22]. □

Corollary 5.2 (small local overlaps in any dimension). *Let C act on $\mathcal H_1\otimes\mathcal H_2\otimes\mathcal H_3$ of arbitrary finite dimensions with $\operatorname{rank}C\le2$. Let $L_i,R_i\subseteq\mathcal H_i$ contain the site-i supports of $\operatorname{ran}C$ and of $\operatorname{ran}C^\dagger$, with inclusion isometries U_i,V_i, and let $M_i=U_i^\dagger V_i$. If $\operatorname{rank}M_i\le4$ for $i=1,2,3$, then $q(C)\ge0$ and $q_\alpha(C)\ge(1-2\alpha)(1-\alpha)^2\|C\|^2$ for $0\le\alpha\le\frac12$.*

Proof. Write $C=UC'V^\dagger$ with $U=\bigotimes U_i$, $V=\bigotimes V_i$ and $\operatorname{rank}C'=\operatorname{rank}C$. Since $\operatorname{vec}C=(U\otimes\bar V)\operatorname{vec}C'$ and $(U_i^\dagger\otimes V_i^{\mathsf T})|\Omega\rangle=\operatorname{vec}M_i$, Proposition 2.1 gives

$$q(C)=\Big\langle\operatorname{vec}C',\ \prod_i\big(I-\tfrac12|\operatorname{vec}M_i\rangle\langle\operatorname{vec}M_i|\big)\operatorname{vec}C'\Big\rangle.$$

Let $M=\sum_{j=1}^k t_j|u_j\rangle\langle v_j|$ be the singular value decomposition of one M_i ($k\le4$, $0<t_j\le1$), and define $W_1|j\rangle=$

$t_j|u_j\rangle$, $W_2|j\rangle = |\bar v_j\rangle$, so that $(W_1\otimes W_2)|\Omega_k\rangle = \operatorname{vec} M$ with $|\Omega_k\rangle = \sum_{j\le k}|jj\rangle$. Expanding,

$$I - \tfrac12|\operatorname{vec} M\rangle\langle\operatorname{vec} M| = (W_1\otimes W_2)\big(I - \tfrac12\Omega_k\big)(W_1\otimes W_2)^\dagger + (I - W_1W_1^\dagger)\otimes W_2W_2^\dagger + I\otimes(I - W_2W_2^\dagger).$$

Multiply out the product over the three sites. In each of the 27 terms, a site carrying one of the last two summands contributes a nonnegative combination of product projectors $|j\rangle\langle j|\otimes|k\rangle\langle k|$, and contracting $\operatorname{vec} C'$ with it compresses C' by a row vector and a column vector at that site, which preserves rank at most two; a site carrying the first summand compresses C' to local dimension $k\le 4$. What remains is the endpoint form on the "active" sites, with local dimensions at most four, which is nonnegative by Theorem 1.1, Theorem 2.3 or Lemma 2.2 according to the number of active sites. The bound for q_α then follows from (7). □

Thus a negative three-copy witness at the endpoint, in any dimension, needs local overlap rank at least five at some site. [18, Theorem 7.2] is the complementary statement that $q\ge 0$ whenever $\operatorname{Tr}(P_iQ_i) = \|M_i\|^2 \le 2$ at a single site i, with P_i, Q_i the projectors onto L_i, R_i and no condition at the other sites; Corollary 5.2 bounds instead the rank of M_i at all three sites and yields the constant of Theorem 1.2.

Corollary 5.3 (full-rank NPT states that survive three copies)**.** *Let $k\ge 3$, $d\ge k$, $\iota:\mathbb{C}^k\to\mathbb{C}^d$ an isometry, $\varrho = (\iota\otimes\iota)\rho_{\alpha_k}^{(k)}(\iota\otimes\iota)^\dagger$ with $\alpha_k = \frac12$ for $k\in\{3,4\}$ and $\alpha_k = \alpha_0$ for $k\ge 5$, and $\tau_\lambda = \lambda\varrho + (1-\lambda)I/d^2$. For $c_k/(d^2+c_k) < \lambda < 1$, with $c_k = k(k-\alpha_k)/(k\alpha_k - 1)$ ($c_3 = 15$, $c_4 = 14$), the state τ_λ has full rank, is NPT, and $\tau_\lambda^{\otimes 3}$ is not one-copy distillable.*

Proof. $(\rho_{\alpha_k}^{(k)})^\Gamma = (I-\alpha_k\Omega)/k(k-\alpha_k)$ has least eigenvalue $-1/c_k$, so the least eigenvalue of τ_λ^Γ is $(1-\lambda)/d^2 - \lambda/c_k$, negative exactly in the stated range. Nondistillability of $\tau_\lambda^{\otimes 3}$ is the compression argument of Remark 3.2(a) with $\rho_{1/2}$ replaced by ϱ, the endpoint input being Theorem 1.1 for $k\in\{3,4\}$ and Theorem 1.3 for $k\ge 5$. □

Corollary 5.4 (robustness)**.** *Let $2\le d_i\le 4$, $0\le\alpha<\frac12$ and $\delta_\alpha = (1-2\alpha)(1-\alpha)^2/\prod_i d_i(d_i-\alpha) > 0$. If σ is a state on $\bigotimes_i(\mathbb{C}^{d_i}\otimes\mathbb{C}^{d_i})$ with $\|\sigma^\Gamma - \bigotimes_i(\rho_\alpha^{(d_i)})^\Gamma\|_{\mathrm{op}} < \delta_\alpha$, then σ is not one-copy distillable; in particular a state σ on $\mathbb{C}^d\otimes\mathbb{C}^d$ with $\|(\sigma^\Gamma)^{\otimes 3} - (\rho_\alpha^\Gamma)^{\otimes 3}\|_{\mathrm{op}} < \delta_\alpha$ is not three-copy distillable. For $1/d<\alpha<\frac12$ the state ρ_α is NPT and NPT is an open condition, so this neighbourhood contains an open set of NPT states. In arbitrary local dimensions the same holds for $0\le\alpha<\alpha_0$ with δ_α replaced by $(1-\alpha/\alpha_0)^3/\prod_i d_i(d_i-\alpha)$.*

Proof. By (6) and Theorem 1.2 (Theorem 1.3 in general dimension), $\langle\psi, \bigotimes_i(\rho_\alpha^{(d_i)})^\Gamma\psi\rangle \ge \delta_\alpha$ for every unit vector $\psi = \operatorname{vec} C$ of Schmidt rank at most two, and the perturbation changes each such expectation by less than δ_α. □

Corollaries 5.1 and 5.3 use only the sign of q_α, i.e. Theorem 1.1 with Remark 3.2(a), or Theorem 1.3; Corollaries 5.2 and 5.4 use the constants of Theorems 1.2 and 1.3.

Remark 5.5. A site of dimension two is free: for $d_i = 2$ the one-site operator $I - \frac12\Omega$ is positive semidefinite with positive semidefinite partial transpose $I - \frac12 F$, hence a separable two-qubit operator [19, 21], i.e. a nonnegative combination of product projectors, so by the compression argument of Corollary 5.2 the endpoint inequality with $d_1 = 2$ is a consequence of the two-site theorem alone.

VI. THE ENDPOINT CERTIFICATE

Every proof of the endpoint inequality that we know, including the one- and two-site theorems, has the same shape. (1) *Lift*: replace C by variables on which "rank at most two" is a polynomial constraint and $q(C)$ a polynomial. (2) *Squares*: exhibit an identity

$$q(C) = \sum_k\langle w_k, G_k w_k\rangle + \sum_j \eta_j e_j \tag{10}$$

on the constraint set, with $G_k\succeq 0$, w_k polynomial vectors in the lifted variables, and e_j polynomials vanishing on the constraint set. (3) Conclude $q\ge 0$. For one site the certificate is a single positive block: with the lift of Lemma 6.1 below, $\operatorname{Tr} C = \langle b, a\rangle$ and $\|C\|^2 = \frac12$, so $q^{(1)}(C) = \frac12(1-|\langle a,b\rangle|^2) = \langle a\otimes b, \frac12(I - F_QF_1)(a\otimes b)\rangle$ with F_QF_1 the swap of the two copies of $\mathcal{V}$ (Lemma 6.2). For two sites, Appendix A is such an identity with a handful of squares. For three sites no closed form is known; the certificates of [1, 2] have hundreds of positive blocks, found by semidefinite programming and then made exact [23]. This section derives the hand lemmas on which they rest.

The lift. Normalise $\|C\|^2 = \frac12$ (the inequality is homogeneous).

Lemma 6.1 (lift)**.** *Let $C\neq 0$ have rank at most two and $\|C\|^2 = \frac12$. Choose an isometry $A\in\mathbb{C}^{d^3\times 2}$ with $\operatorname{ran} C\subseteq\operatorname{ran} A$, and put $B = 2C^\dagger A$. Then*

$$A^\dagger A = I_2, \qquad C = \tfrac12 AB^\dagger, \qquad \|B\|^2 = 2, \tag{11}$$

and nothing else is imposed on B.

Proof. $AA^\dagger$ projects onto $\operatorname{ran} A\supseteq\operatorname{ran} C$, so $\frac12 AB^\dagger = AA^\dagger C = C$, and $\|B\|^2 = 4\operatorname{Tr}(C^\dagger AA^\dagger C) = 4\|C\|^2$. □

Let $\mathcal{V} = \mathcal{H}\otimes\mathbb{C}^2$ (an auxiliary qubit) and

$$a = \tfrac{1}{\sqrt2}\sum_{j,\mu} A_{j\mu}|j\rangle|\mu\rangle, \qquad b = \tfrac{1}{\sqrt2}\sum_{j,\mu} B_{j\mu}|j\rangle|\mu\rangle \qquad \in\mathcal{V}. \tag{12}$$

Then $\|a\| = \|b\| = 1$, $C_{jk} = \sum_\mu a_{j\mu}\bar b_{k\mu}$, and the isometry condition says that the qubit marginal of a is maximally mixed, $\operatorname{Tr}_{\mathcal{H}}|a\rangle\langle a| = \frac12 I_2$. "Rank at most two" has become a *linear* condition on a marginal of a; the constraint set is $\mathcal{S} = \{(a,b) : \operatorname{Tr}_{\mathcal{H}}|a\rangle\langle a| = \frac12 I_2,\ \|b\| = 1\}$.

Lemma 6.2 (swap form)**.** *On $\mathcal{V}\otimes\mathcal{V}$ let F_i swap the site-i factors and F_Q the qubit factors of the two copies. For every $S\subseteq\{1,2,3\}$, $\langle a\otimes b, F_Q\prod_{i\in S}F_i\,(a\otimes b)\rangle = \|\operatorname{Tr}_S C\|^2$; hence*

$$q(C)=\langle a\otimes b, Z\,(a\otimes b)\rangle,\qquad Z=F_Q\prod_{i=1}^{3}\big(I-\tfrac{1}{2}F_i\big). \quad (13)$$

Proof. With s,t running over the indices of the sites in S, j,k over the remaining sites and μ,ν over the qubit, $(\operatorname{Tr}_S C)_{jk}=\frac{1}{2}\sum_{s,\mu}A_{sj,\mu}\bar{B}_{sk,\mu}$, so $\|\operatorname{Tr}_S C\|^2=\frac{1}{4}\sum A_{sj,\mu}\bar{B}_{sk,\mu}\bar{A}_{tj,\nu}B_{tk,\nu}$, which is the left side: F_Q exchanges the qubit indices and F_i the site-i indices between the two copies. Expand with (2). □

Z is indefinite (F_Q and the F_i commute and have eigenvalues ±1); Theorem 1.1 says that Z is nonnegative on the product vectors $a\otimes b$ with $(a,b)\in\mathcal{S}$.

Moments. Take four replicas of $\mathcal{V}$, numbered $0,1,2,3$, write $P_a=|a\rangle\langle a|$, $P_b=|b\rangle\langle b|$, and consider

$$Y_{31}=P_a^{\otimes3}\otimes P_b,\quad Y_{22}=P_a^{\otimes2}\otimes P_b^{\otimes2},\quad Y_{13}=P_b^{\otimes3}\otimes P_a, \quad (14)$$

whose entries are the monomials of bidegree $(3,1)$, $(2,2)$, $(1,3)$ in (P_a,P_b); by Lemma 6.2, $q(C)=\operatorname{Tr}[(Z_{03}\otimes I_{\{1,2\}})Y_{31}]$ with Z acting on replicas 0 and 3. For $f\in\{31,22,13\}$ and a set T of replicas, the partial transpose $Y_f^{\Gamma_T}$ of the whole factor $\mathcal{V}$ in the replicas of T is again a product of rank-one projectors (the vector a or b carried by each replica in T replaced by its complex conjugate), hence positive semidefinite, and $\operatorname{Tr}[GY_f^{\Gamma_T}]$ with $G\succeq0$ is a sum of squared moduli of polynomials in $(a,\bar a,b,\bar b)$; we call a pair (f,T) a *word*. The certificates use thirteen words and one *coupled* block [1, 2]: with $u=a\otimes a\otimes b\otimes\bar a$ and $v=b\otimes b\otimes a\otimes\bar b$ in $\mathcal{V}^{\otimes4}$,

$$\begin{pmatrix}|u\rangle\langle u| & |u\rangle\langle v|\\ |v\rangle\langle u| & |v\rangle\langle v|\end{pmatrix}=|u\oplus v\rangle\langle u\oplus v|\succeq0 \quad (15)$$

is a linear image of the moments: its diagonal blocks are Y_{31} and Y_{13} with replicas 2 and 3 exchanged and replica 3 transposed, and its off-diagonal block is the entrywise rearrangement $(Y_{22})_{i_0i_1i_2j_3;\,i_3j_2j_0j_1}$ of Y_{22}, which equals $u_i\bar v_j$ [1, Sec. 3.2]. A square $\langle u\oplus v, G(u\oplus v)\rangle$ contains cross terms between an a-heavy and a b-heavy polynomial. On $\mathcal{S}$ the moments satisfy linear relations: the qubit marginal of a is $\frac{1}{2}I_2$ at every replica carrying a, and the three moments share their common three-replica marginals.

Symmetry. q, $\mathcal{S}$ and all of the above are invariant under the group $\mathcal{G}=U(d)^{\times3}\times U(2)\times S_3\times\{1,\mathrm{conj}\}$ acting on $\mathcal{V}$ (an element g of the unitary part replaces C by $gCg^\dagger$ with $g\in U(d)^{\times3}$; S_3 permutes the sites; conj conjugates (a,b)). By Schur–Weyl duality [24, Lect. 6], moments averaged over $\mathcal{G}$ lie in the span of products of permutation operators of the four replicas, one factor per site and one for the qubit, cut down by the bosonic support of the Y_f, by S_3 and by conjugation. Let ξ be the coordinate vector of the averaged moments in a fixed rational basis of this span, with one *sector* for each of the three moments (dimensions in Table I). The objective is then a rational linear functional $\xi\mapsto c^{\mathsf T}\xi$, the relations an integer matrix K with $K\xi=0$, and the partially transposed averaged moments and the averaged coupled block decompose into isotypic *blocks*, onto which rational linear maps $\widehat{Y}_t$ ($t=1,\dots,716$ for $d=3$, $1,\dots,1239$ for $d=4$; 64, respectively 111, of them from the coupled block) compress them. Concretely,

$$\widehat{Y}_t(\xi)=N_t^{\mathsf T}J_t^{\mathsf T}\,\mathcal{M}_t(\xi)\,J_tN_t, \quad (16)$$

where $\mathcal{M}_t(\xi)$ is the partially transposed averaged moment, or the averaged coupled block, of the word of t; J_t is a stored rational embedding of the carrier of an isotypic block into $\mathcal{V}^{\otimes4}$ (into $\mathcal{V}^{\otimes4}\oplus\mathcal{V}^{\otimes4}$ for the coupled block), so that $J_t^{\mathsf T}\mathcal{M}_tJ_t$ is that block in a rational basis; and N_t is a stored rational frame selecting a subspace of it. The frames were chosen to annihilate the known zeros, which is why every $\widehat{Y}_t(\xi)$ can vanish at a zero (Remark 4.5(c)) although the moments have trace one.

Lemma 6.3 (averaging)**.** *Let $(a,b)\in\mathcal{S}$ and let ξ be the coordinate vector of the moments (14) averaged over $\mathcal{G}$. Then $c^{\mathsf T}\xi=q(C)$, $K\xi=0$, and $\widehat{Y}_t(\xi)\succeq0$ for every t.*

Proof. $q(C)=\operatorname{Tr}[(Z_{03}\otimes I_{\{1,2\}})Y_{31}]$, and Z commutes with $g\otimes g$ for $g\in U(d)^{\times3}\times U(2)$, is invariant under site permutations and is real, so the objective is unchanged by averaging. The relations are linear in the moments and $\mathcal{G}$-covariant, so they hold for the average because they hold for each term. Each $Y_f^{\Gamma_T}$ and the coupled block (15) are positive semidefinite for every (a,b), and $\mathcal{G}$ acts on them by conjugation with g on the untransposed and $\bar g$ on the transposed replicas—on both halves of (15) by $g\otimes g\otimes g\otimes\bar g$—and antiunitarily for conj; so the averaged operators and their compressions (16) are positive semidefinite. By Schur–Weyl duality the averaged moments lie in the span of products of permutation operators, in which ξ is the coordinate vector. □

Theorem 6.4 (four-replica certificate; $d=3$: [1], $d=4$: [2])**.** *For $d=3$ and for $d=4$ there are rational symmetric positive semidefinite matrices G_t ($t=1,\dots,716$, respectively $1,\dots,1239$; positive definite for $d=4$) and a rational vector η such that, identically in ξ,*

$$c^{\mathsf T}\xi=\sum_t\operatorname{Tr}\big[G_t\,\widehat{Y}_t(\xi)\big]+\eta^{\mathsf T}K_0\xi, \quad (17)$$

where c, K and $\widehat{Y}_t$ are the objects of Lemma 6.3 in the stored coordinate basis of ξ, and K_0 consists of 272 rows of K.

Proof of Theorem 1.1. Let $C\neq0$ have rank at most two; normalise, lift by Lemma 6.1, and let ξ be as in Lemma 6.3. Then $c^{\mathsf T}\xi=q(C)$, $K_0\xi=0$ and $\widehat{Y}_t(\xi)\succeq0$, so each term $\operatorname{Tr}[G_t\widehat{Y}_t(\xi)]=\operatorname{Tr}[G_t^{1/2}\widehat{Y}_t(\xi)G_t^{1/2}]$ in (17) is nonnegative and $q(C)\geq0$. □

	$d=3$ [1]	$d=4$ [2]
$\dim \mathcal{V}$	54	128
coordinates of ξ	2374	2741
sectors $31, 22, 13$	$577, 1220, 577$	$667, 1407, 667$
relations K: rows, rank	2912, 272	2912, 272
blocks G_t (coupled)	716 (64)	1239 (111)
largest block order	46	43
positivity test	char. polynomial	diag. dominance
exact replay	≈ 4 min	≈ 7 min

TABLE I. The two four-replica certificates. "Sectors" are the dimensions of the three blocks of ξ; "coupled" counts the blocks from the coupled block (15); replay times are wall-clock times of the upstream verifiers on two cores.

Remark 6.5. The averaged moment set is an *outer* relaxation of the physical one; the proof never needs that every feasible ξ comes from some (a, b). The $d = 4$ certificate differs from the $d = 3$ one only in that the four-replica antisymmetriser of $\mathbb{C}^4$ has rank one instead of vanishing, which enlarges the sectors (Table I); the same thirteen words, coupled block, relations and objective are used. Since the case $d = 3$ is also a corollary of the case $d = 4$ by Lemma 2.4(b), the qutrit endpoint rests on two independently computed certificates of the same construction, with different sector bases, exact positivity tests and verifier programs. Their agreement does not exclude a shared error in the reduction of this section, which is why Lemmas 6.1–6.3 are proved here and the control program of Sec. VIII evaluates the stored data on actual matrices.

VII. EVERY DIMENSION UP TO α_0

The certificates of Sec. VI are tied to a dimension: the sector bases change with d (Table I), and the Gram matrices of the permutation operators are polynomials in d whose leading terms differ from block to block. To obtain a certificate that is uniform in d we change the lift, so that the free variables are orthonormal frames rather than vectors with a prescribed marginal; replace the moment matrix by a 2×2 determinant, which is a quadratic form in four replicas of $\mathcal{H}$ itself; and work in the algebra generated by the partially transposed permutation operators at one site, whose structure constants are polynomials in d. Positivity for all sufficiently large d is then a condition on finitely many leading coefficients, and an isometric embedding carries it to every dimension. The price is the value of the parameter: the method proves $q_{\alpha_0} \geq 0$ for

$$\alpha_0 = \tfrac{37}{80} = 0.4625, \tag{18}$$

not for $\alpha = \frac{1}{2}$.

Singular frames and the determinant. Let $C = s_0 u_0 v_0^\dagger + s_1 u_1 v_1^\dagger$ be a singular value decomposition, $s_j \geq 0$, (u_0, u_1) and (v_0, v_1) orthonormal pairs in $\mathcal{H}$ (a rank-one C has $s_1 = 0$). For real weights $w = (w_S)_{S\subseteq\{1,2,3\}}$ put $q_w(C) = \sum_S w_S \| \operatorname{Tr}_S C\|^2$, $D_j = u_j v_j^\dagger$ and

$$(H_w)_{jk} = \sum_S w_S \,\langle \operatorname{Tr}_S D_j, \operatorname{Tr}_S D_k\rangle, \qquad j,k \in \{0,1\}. \tag{19}$$

Then $q_w(s_0 D_0 + s_1 D_1) = s^{\mathsf{T}} H_w s$, and since $c_0 D_0 + c_1 D_1$ has rank at most two for all complex c, $q_w \geq 0$ on matrices of rank at most two if and only if $H_w \succeq 0$ for every pair of frames. For $w_S = (-\alpha)^{|S|}$ the diagonal entries are $q_\alpha(u_j v_j^\dagger) = \langle u_j \otimes v_j, \prod_i (I - \alpha F_i)\, u_j \otimes v_j\rangle \geq (1-\alpha)^3$ by the rank-one identity of Sec. IV (the three swaps commute and have eigenvalues ± 1), so for $0 \leq \alpha < 1$,

$$q_\alpha \geq 0 \text{ on rank} \leq 2 \iff \det H_\alpha \geq 0 \text{ on all frames.} \tag{20}$$

The determinant is a quadratic form in four replicas. Number the replicas $(u_0, v_0, u_1, v_1) = (0, 1, 2, 3)$, let $r = u_0 \otimes v_0 \otimes u_1 \otimes v_1 \in \mathcal{H}^{\otimes 4}$, and for $p \in S_4$ let P_p permute the replicas at one site, $(P_p)_{y,x} = \prod_j \delta(y_{p(j)}, x_j)$ (input position j goes to output position $p(j)$). In $\langle r, \bigotimes_i P_{p_i}\, r\rangle$ the site-i index of replica j in r is contracted with that of replica $p_i(j)$ in $\bar{r}$; when every replica is a product vector over the sites this is $\prod_i \prod_j \langle r^{(p_i(j))}, r^{(j)}\rangle_i$. With $\sigma_e = (01)^e$, $\sigma'_f = (23)^f$ for $e, f \in \{0,1\}$, Lemma 6.2 gives $(H_w)_{00}(H_w)_{11} = \langle r, \sum_{S,R} w_S w_R \bigotimes_i P_{\sigma_{e_i}\sigma'_{f_i}}\, r\rangle$ with $e_i = [i \in S]$, $f_i = [i \in R]$, and the same bookkeeping for $(H_w)_{01}(H_w)_{10}$ gives

$$\begin{aligned} \det H_w &= \langle r, K(w)\, r\rangle, \\ K(w) &= \sum_{S,R} w_S w_R \Big(\bigotimes_i P_{\sigma_{e_i}\sigma'_{f_i}} - \bigotimes_i P_{\kappa_{e_i f_i}} \Big), \end{aligned} \tag{21}$$

where $\kappa_{00} = (02)(13)$, $\kappa_{11} = (01)(23)$, $\kappa_{10} = (0231)$ and $\kappa_{01} = (0132)$ (cycle notation). $K(w)$ is a combination of 126 products of permutation operators with coefficients quadratic in w.

Squares, the ideal and the hand inputs. Three kinds of terms have nonnegative expectation in every product vector r of orthonormal frames. (i) For a set τ of replicas let Γ_τ transpose the replicas in τ at all three sites. If $\Gamma_\tau(Y) \succeq 0$ then $\langle r, Y r\rangle = \langle r^{*\tau}, \Gamma_\tau(Y)\, r^{*\tau}\rangle \geq 0$, where $r^{*\tau}$ conjugates the vectors in τ; complementary sets give the same condition, so there are eight: $\tau = \emptyset$, the four singletons and $\{0,1\}, \{0,2\}, \{0,3\}$. (ii) In a product $\bigotimes_i P_{p_i}$ with $p_i(2) = 0$ at every site, all site indices of replica 2 are contracted with those of replica 0, so its expectation contains the factor $\langle u_0, u_1\rangle = 0$; the same holds for $p_i(0) = 2$, $p_i(1) = 3$ and $p_i(3) = 1$. These 819 products span a subspace $\mathcal{I}$ of forms that vanish on orthonormal frames. (iii) If a and a' are weight vectors such that q_a and $q_{a'}$ are nonnegative on matrices of rank at most two in all dimensions, then $H_a, H_{a'} \succeq 0$ on all frames and, with J the exchange of the two rows,

$$T_\pm(a, a') = \operatorname{Tr}\big[H_a\,(J H_{a'}^{\mathsf{T}} J \circ \Sigma_\pm)\big] \geq 0, \;\; \Sigma_\pm = \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}, \tag{22}$$

by the Schur product theorem; each $T_{\pm}(a,a')$ is again a combination of permutation products, and $T_{-}(a,a) = 2\det H_a$. The certificate uses a single hand input of this kind: the weight vector a_B of Theorem B.1 ($a_\emptyset = \frac{17}{8}$, $a_S = -1, \frac{1}{2}, -\frac{5}{16}$ for $|S| = 1,2,3$), for which $q_{a_B}(C) \geq \frac{1}{4}(s_1 - s_2)^2 \geq 0$ in every dimension, so that

$$T_B := \det H_{a_B} = \langle r, K(a_B)\, r\rangle \geq 0 \tag{23}$$

on all frames. The one- and two-site theorems in the same form (Lemma 2.2, Theorem 2.3 and their versions with a spectator site) were offered to the search as well and are not needed (Remark 7.3(b)).

The algebra at one site. Fix a site, its dimension d and a set τ of replicas. The 24 operators $D_p = \Gamma_\tau(P_p)$ on $(\mathbb{C}^d)^{\otimes 4}$ are the diagrams of the walled Brauer algebra $B_{4-|\tau|,|\tau|}(d)$ [25–27] (positive maps built from this algebra are studied in [28]): $D_p D_q = d^{\,\ell(p,q)} D_{p\circ_\tau q}$, where $p \circ_\tau q$ and the number $\ell(p,q)$ of closed loops are read off from the concatenated diagrams. Its irreducible modules for generic d [29] are indexed by the number $k \leq \min(|\tau|, 4-|\tau|)$ of arcs and a pair of partitions $\lambda \vdash 4-|\tau|-k$, $\mu \vdash |\tau| - k$, and have dimensions $\binom{4-|\tau|}{k}\binom{|\tau|}{k} k!\, f^\lambda f^\mu$: $1,3,2,3,1$ for $\tau = \emptyset$ (the irreducible representations of S_4), $1,2,1,3,3$ for $|\tau| = 1$ and $1,1,1,1,4,2$ for $|\tau| = 2$, with $\sum_r (\dim W_r)^2 = 24$ in each case. We realise each module W_r on the span of the half-diagrams with k arcs tensored with the polytabloid bases of the Specht modules of λ and μ; the action of D_p and the contravariant bilinear form are defined by concatenation, with a factor d per closed loop, a concatenation that creates more than k arcs acting as zero, and the invariant form of the Specht modules on the polytabloid factor. This yields matrices $\rho_r(p;d)$ and symmetric $G_r(d)$ with polynomial entries of degree at most k, such that

$$\rho_r(p)\rho_r(q) = d^{\,\ell(p,q)}\rho_r(p\circ_\tau q), \qquad \rho_r(p)^{\mathsf T} G_r = G_r\, \rho_r(p^*), \tag{24}$$

where p^* labels the adjoint diagram ($D_p^\dagger = D_{p^*}$).

Lemma 7.1. *For all sufficiently large d: every $G_r(d)$ is positive definite, the map $\Phi : D_p \mapsto (\rho_r(p;d))_r$ is injective, and hence Φ is a $*$-isomorphism of the algebra $A_\tau(d)$ spanned by the D_p onto its image in $\bigoplus_r \mathrm{End}(W_r)$, where W_r carries the inner product G_r. Consequently a self-adjoint element $Y = Y^* = \sum_p y_p \bigotimes_{i=1}^3 D_{p_i}$ of $A_\tau(d)^{\otimes 3}$ is positive semidefinite on $(\mathbb{C}^d)^{\otimes 12}$ if and only if for every triple $r = (r_1, r_2, r_3)$ the symmetric matrix*

$$B_{\tau,r}(d) = \Big(\bigotimes_j G_{r_j}\Big) \sum_p y_p \bigotimes_j \rho_{r_j}(p_j) \tag{25}$$

is positive semidefinite.

Proof. The leading coefficient of $G_r(d)$ is positive definite and Φ is injective at $d = 5$ (both checked exactly, Sec. VIII); a nonzero 24×24 minor is a polynomial in d with finitely many roots. By (24), Φ is a $*$-homomorphism when $\mathrm{End}(W_r)$ carries the adjoint $M \mapsto G_r^{-1} M^{\mathsf T} G_r$, and $A_\tau(d)$ is a $*$-subalgebra of $\mathrm{End}((\mathbb{C}^d)^{\otimes 4})$. If $Y = Y^*$ and $\Phi(Y) \succeq 0$, then $\Phi(Y)^{1/2}$ is a polynomial in $\Phi(Y)$, hence equals $\Phi(V)$ for some $V = V^*$ in the algebra, and $Y = V^2$ by injectivity; so $Y \succeq 0$. The converse holds because Φ is a $*$-homomorphism, and the three-site statement follows in the same way for $A_\tau(d)^{\otimes 3}$, whose blocks are the tensor products of the W_r with the inner products $\bigotimes G_{r_j}$. The proof of Theorem 1.3 uses the direction "blocks positive $\Rightarrow$ Y positive". □

Large d. Let k_r be the largest degree in d of an entry of any $\rho_r(p;d)$ and g_r that of $G_r(d)$. With $t = 1/d$, $\bar B_{\tau,r}(t) = t^{\sum_j (k_{r_j} + g_{r_j})} B_{\tau,r}(1/t)$ is a polynomial matrix in t whose value at $t = 0$ is built from the leading coefficients of the ρ_{r_j} and G_{r_j} alone. If $\bar B_{\tau,r}(0) \succ 0$ then $B_{\tau,r}(d) \succ 0$ for all sufficiently large d, by continuity. So a finite list of strict inequalities between rational matrices implies positivity of $\Gamma_\tau(Y)$ in all sufficiently large dimensions.

Descent. By Lemma 2.4(b), a matrix of rank at most two on $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$ of arbitrary dimensions has an isometric image on $(\mathbb{C}^D)^{\otimes 3}$ for every $D \geq \max_i d_i$, of the same rank and with the same q_α; it therefore suffices to prove $q_{\alpha_0} \geq 0$ for all sufficiently large D. For $0 \leq \alpha \leq \alpha_0$ the decomposition $I - \alpha\Omega = \frac{\alpha}{\alpha_0}(I - \alpha_0\Omega) + (1 - \frac{\alpha}{\alpha_0})I$ at each site and the compression argument of Remark 3.2(a) give $q_\alpha(C) \geq (1 - \alpha/\alpha_0)^3 \|C\|^2$: the terms with one or two active sites are one- and two-site forms at the parameter $\alpha_0 \leq \frac{1}{2}$, nonnegative by Lemma 2.2 and Theorem 2.3 with the same decomposition at $\frac{1}{2}$, and the term with three active sites is $q_{\alpha_0} \geq 0$.

Theorem 7.2 (uniform certificate)**.** *With α_0 as in (18), there are rational numbers $y_{\tau,p}$ for the eight sets τ and $p \in S_4^3$, a rational $\mu \geq 0$ and a form $E \in \mathcal{I}$ such that*

$$K(\alpha_0) = \sum_\tau Y_\tau + E + \mu\, T_B, \qquad Y_\tau = \sum_p y_{\tau,p} \bigotimes_i P_{p_i}, \tag{26}$$

as an identity between the 13 824 coefficients, with $Y_\tau = Y_\tau^*$, and $\bar B_{\tau,r}(0) - \varepsilon I \succ 0$ with

$$\varepsilon = 243444/653071328197 \approx 3.7 \times 10^{-7}$$

for $\tau \in \{\emptyset, \{0\}, \{0,1\}, \{0,2\}, \{0,3\}\}$ and every unordered triple r of blocks (238 matrices of order at most 64, Table II); $Y_{\{j\}}$ is the relabelling of $Y_{\{0\}}$ by the element of V_4 taking 0 to j, and the 105 leading blocks of these three sets are positive definite as well. Moreover, for each of the eight sets τ, the matrices $\rho_r(p;d)$ and $G_r(d)$ satisfy (24), $\sum_r (\dim W_r)^2 = 24$, the leading coefficient of every G_r is positive definite, and Φ has rank 24 at $d = 5$.

Proof of Theorem 1.3. Let D be large enough for Lemma 7.1 and for $B_{\tau,r}(D) \succ 0$, and let C on $(\mathbb{C}^D)^{\otimes 3}$ have rank at most two, with frames u_j, v_j and r as above. By Theorem 7.2 and Lemma 7.1, $\Gamma_\tau(Y_\tau) \succeq 0$, so $\langle r, Y_\tau r\rangle \geq 0$ by (i); $\langle r, Er\rangle = 0$ by (ii); $\mu\langle r, T_B r\rangle \geq 0$ by (iii). Hence $\det H_{\alpha_0} = \langle r, K(\alpha_0) r\rangle \geq 0$, and (20)

τ	block orders	$k_r = g_r$	triples	max. order
$\emptyset$	$1,3,2,3,1$	0	35	27
$\{j\}$	$1,2,1,3,3$	$0,0,0,1,1$	35	27
$\{0,j\}$	$1,1,1,1,4,2$	$0,0,0,0,1,2$	56 each	64

TABLE II. The dimension-uniform certificate: the one-site blocks for each class of transpose sets, the degrees in d of their representation and Gram matrices, and the number of unordered triples of blocks tested at leading order (238 in all). The identity (26) has 819 ideal monomials in 40 orbits.

gives $q_{\alpha_0}(C) \geq 0$. Descent extends this to every dimension and to $0 \leq \alpha \leq \alpha_0$, and Proposition 2.1 translates it into three-copy undistillability. □

Remark 7.3. (a) The symmetries of K (site permutations, the relabellings $u_j \leftrightarrow v_j$ and $0 \leftrightarrow 1$ of the dyads, which form the Klein group V_4 acting on the replicas, the adjoint and complex conjugation) are imposed on the certificate: $Y_{\{j\}}$ is the relabelling of $Y_{\{0\}}$, the other Y_τ are invariant, and only unordered triples of blocks need to be listed, since $\Gamma_{g\tau}(gYg^{-1}) = g\,\Gamma_\tau(Y)\,g^{-1}$ for a relabelling g and Y_τ is invariant under site permutations. The identity (26) then has one equation per orbit of monomials outside $\mathcal{I}$ (499 equations) and 3 602 unknowns. (b) The hand input enters with multiplier $\mu = 796570481880293/2^{53} \approx 0.0884$. In the floating-point search (a bisection in α over the leading-order blocks, with the solver reporting reduced accuracy near the boundary), the squares and the subspace $\mathcal{I}$ alone are feasible up to $\alpha \approx 0.4603$; the one- and two-site theorems in the form (22) (the 24 distinct symmetrised products of this kind were offered) are still infeasible at 0.4625; with T_B, with or without those products, the search is feasible at 0.4625 and infeasible at about 0.4627. So Theorem B.1 is what carries the certificate from 0.4603 to α_0, and we stopped at the round value 37/80 to keep the margin ε comfortable. (c) The leading-order condition $\bar{B}(0) \succ 0$ is sufficient, not necessary, for positivity at all large d; the parameter α_0 is what this condition yields with four replicas and the hand input T_B. We do not know the supremum of the parameters attainable by four-replica certificates, nor whether a dimension-uniform proof of the endpoint $\alpha = \frac{1}{2}$ exists in this framework (Sec. IX).

VIII. WHAT THE COMPUTER CHECKS

Following the convention of [30], we list what rests on a computation. Theorem 6.4, in each of the two dimensions, is the only external computational theorem required for Theorems 1.1 and 1.2; its statement includes that the stored basis, c, K and $\widehat{Y}_t$ are the objects of Sec. VI. Theorem 7.2 is the only computational input of Theorem 1.3. The numerical observations of Remark 4.5(a),(b) and the control calculations below are separate and enter no proof. The verifiers of [1] and [2] establish the following finite statements in exact integer and rational arithmetic: (1)–(2) the stored bases of the three sectors are bases (their number equals the character-sum dimension and an integer minor is nonzero modulo a prime) and intertwine the 24 replica permutations and the partial transposes; (3) the objective c and the relation matrix K are recomputed from the literal partial-trace definitions of Sec. VI and agree with the stored ones; (4) every map $\widehat{Y}_t$ is rebuilt from the stored sector bases and agrees with the stored one; (5) every G_t is positive semidefinite, for $d = 3$ by the sign pattern of the characteristic polynomial of a scaled integer matrix, for $d = 4$ by a rational congruence to a strictly diagonally dominant matrix with margin $> 9999992/10^7$ in each of its 4231 rows; (6) the identity (17) holds on every basis vector: all 2374, respectively 2741, rational residuals vanish. We replayed both verifiers; both pass, and the $d = 3$ verifier rejects its four supplied negative controls. Neither certificate has appeared in a refereed journal at the time of writing; the $d = 3$ endpoint is therefore stated with both routes (directly, and by compression from $d = 4$), and the replay receipts are in [20].

For Theorem 7.2 the data are our own and small (the $y_{\tau,p}$ on orbit representatives, μ, α_0 and ε: about 200 kB of JSON), and the verifier recomputes everything else from the definitions of Sec. VII in exact rational arithmetic: (7) the diagram calculus agrees with the explicit operators on $(\mathbb{C}^d)^{\otimes 4}$ for $d = 2, 3$ (all 105 diagrams, all products, traces, adjoints and transposes), and for each of the eight sets τ the block matrices $\rho_r(p;d)$ and $G_r(d)$ constructed from the half-diagrams and polytabloids satisfy (24) as polynomial identities (576 products per block), have $\sum_r (\dim W_r)^2 = 24$, leading Gram coefficients that are positive definite, and a block map of rank 24 at $d = 5$; (8) $K(\alpha_0)$, the subspace $\mathcal{I}$ and $T_B = K(a_B)$ are rebuilt from (21) and (ii), and the identity (26) holds coefficient by coefficient, with $Y_\tau = Y_\tau^*$ and $\mu \geq 0$; (9) every leading block of the five sets satisfies $\bar{B}_{\tau,r}(0) - \varepsilon I \succ 0$, and the 105 leading blocks of the three relabelled sets are positive definite, by exact LDL^{T} factorisations with positive pivots (ε is a margin in the stored half-diagram and polytabloid bases, not an invariant of the blocks). The verifier runs in about 70 s on one core.

The verifiers check the stored data, not the correspondence between an actual matrix C and its coordinates (Lemmas 6.1–6.3). As an independent control, a short floating-point program builds the lift of a given 64×64 matrix C, evaluates the averaged moments from the literal permutation operators, and evaluates the stored $d = 4$ data on the resulting ξ: for random non-normal C it reproduces $c^{\mathsf{T}}\xi = q(C)$, $K\xi = 0$, $\widehat{Y}_t(\xi) \succeq 0$ and (17) to machine precision, and at the zero $2^{-1/2}P_E \otimes |\varphi\rangle\langle\chi|$ every $\widehat{Y}_t(\xi)$ vanishes. A second script checks the hand identities of this paper, including (21), (22) and the identities of Appendix B, symbolically or on random instances. Neither is part of the proof.

The package [20] contains these scripts, the uniform certificate of Theorem 7.2 with its verifier and receipt,

the census of Remark 4.5, scripts that fetch the two certificates at pinned commits and check a SHA-256 digest of every tracked file before running the upstream verifiers, the receipts of our runs, and a table identifying the objects of Sec. VI with the stored data (basis and scaling of ξ, normalisation of c, the literal relations and the selected rows K_0, the embeddings J_t, frames N_t and the rearrangement of the coupled block for every t); the certificate data themselves are published [1, 2] and are not redistributed (the repository of [2] states no licence). Not verified by any program: the hand proofs, the interpreters and the hardware. The floating-point semidefinite programs that discovered the certificates are not part of the proof and are not needed to verify it; the one that found Theorem 7.2 (a bisection in α over the leading-order blocks, 3 602 unknowns) is included for reproducibility.

Independently of the certificates above, Theorems 1.1 and 1.2 for $d = 3$, Proposition 4.2(a) for $d = 3$ and Corollary 5.3 for $k = 3$ have been formalized in Lean 4 and checked by Lean's kernel, the qutrit endpoint inequality from an exact certificate of PSI's own (see the AI disclosure).

IX. WHAT REMAINS

Higher local dimension. In local dimension $d \geq 5$ the inequality $q_\alpha \geq 0$ on rank two is open for $\alpha_0 < \alpha \leq \frac{1}{2}$: Theorem 1.3 gives $0 \leq \alpha \leq \alpha_0$ in every dimension, Corollary 5.2 covers all matrices whose local range–corange overlaps have rank at most four at the endpoint, and the normal and several non-normal classes are covered at the endpoint in every dimension by [18]. The two kinds of certificate are complementary. The endpoint certificates of Sec. VI do not induct in d: the compressions $\widehat{Y}_t$ depend on d through the Gram matrices of the permutation operators, and a new isotypic component (the four-replica antisymmetriser of $\mathbb{C}^4$) appears at $d = 4$, so a certificate in one dimension is not one in a larger dimension, although a certificate in dimension d' compresses to one in every $d \leq d'$. The uniform certificate of Sec. VII treats d as a parameter but tests positivity only at leading order in $1/d$, which is one reason it stops at α_0; whether a dimension-uniform four-replica certificate can reach $\alpha = \frac{1}{2}$, by using the lower-order terms or more hand inputs, we do not know.

More copies. Nothing inducts in the number of copies either. Whether $\rho_{1/2}$ (for some $d \geq 3$) is undistillable from every finite number of copies—the NPT bound-entanglement conjecture—is untouched. Two remarks apply at every count: the convex decomposition of Remark 3.2(a) applies to every number of copies, so the endpoint statement at a given count implies the sign on the whole interval; and the endpoint zero set contains, for every number r of copies, the rank-two kernels $\beta(x, y)$ on r qubits, of alternating algebraic type, so that any inductive argument has to be exactly tight on them.

A human proof. The certificates give the positive blocks only as tables. A proof a human can read would be the same identity (10) with a short, structured choice of the blocks: squares of degree four in the lifted pair (a, b), tight on the three families of Sec. IV (conjecturally the whole zero set, of dimension 36 for $d = 3$) and on both algebraic types of zero, with the rank-two condition entering only through the marginal constraint on a. The one- and two-site instances (Sec. VI and Appendix A) show what such a closed form looks like one and two copies earlier.

Appendix A: The two-site inequality as a sum of squares

We assemble Theorem 2.3 for rank two as an identity between nonnegative terms, the two-site instance of (10), from the ingredients of [16, §§6–8]: the balanced factorisation [16, Lemma 7.1], the Parker–Fillmore lemma [16, Lemma 6.1], the rank-one identity of [13, Prop. 1] in the form [16, Lemma 6.4], and the crossed polarisation [16, Cor. 6.7] of a four-state identity of [31].

Let $C = XY^\dagger$ have rank at most two on $\mathcal{H}_L \otimes \mathcal{H}_R$, with *balanced* factors $X = [x_1\ x_2]$, $Y = [y_1\ y_2]$: $X^\dagger X = Y^\dagger Y = \left(\begin{smallmatrix} n_1 & \omega \\ \bar{\omega} & n_2 \end{smallmatrix}\right)$ (take $X = U\Sigma^{1/2}W$, $Y = V\Sigma^{1/2}W$ from a singular value decomposition $C = U\Sigma V^\dagger$, $W \in U(2)$ free). Choose W so that $y_1^\dagger x_1 = y_2^\dagger x_2 =: t$: conjugation by W rotates the complex Pauli coefficient vector of the traceless part of $Y^\dagger X$ by a real rotation, and the real and imaginary parts of that vector span at most a plane, which can be rotated into the plane orthogonal to the third axis. On two copies of $\mathcal{H}_L \otimes \mathcal{H}_R$ let $J = \frac{1}{4}(I - F_L)(I - F_R)$ and $\Lambda(M) = (\mathrm{Tr}_L\, M, \mathrm{Tr}_R\, M)$ with the sum inner product, and put $s_X = 4\|J(x_1 \otimes x_2)\|^2$, $s_Y = 4\|J(y_1 \otimes y_2)\|^2$, $s_j = 4\|J(x_j \otimes y_j)\|^2$, $\Delta = x_2 x_1^\dagger - y_2 y_1^\dagger$. Expanding $4J$ into its four swap words and contracting as in Proposition 2.1 gives, for arbitrary vectors,

$$\|\Lambda(xy^\dagger)\|^2 = \|x\|^2\|y\|^2 + |\langle y, x\rangle|^2 - 4\|J(x \otimes y)\|^2, \quad \text{(A1)}$$

$$\langle \Lambda(x_1 y_1^\dagger), \Lambda(x_2 y_2^\dagger)\rangle = \langle \Lambda(y_2 y_1^\dagger), \Lambda(x_2 x_1^\dagger)\rangle \quad \text{(A2)}$$

(in (A2) each traced subsystem pairs with its complement, which exchanges the roles of the two frames). Under balance and equal overlaps, (A1) gives $\|\Lambda(x_j y_j^\dagger)\|^2 = n_j^2 + |t|^2 - s_j$, $\|\Lambda(x_2 x_1^\dagger)\|^2 = n_1 n_2 + |\omega|^2 - s_X$ and $\|\Lambda(y_2 y_1^\dagger)\|^2 = n_1 n_2 + |\omega|^2 - s_Y$. Substituting into $Q_{L|R}(C) = n_1^2 + n_2^2 + 2|\omega|^2 - \frac{1}{2}\sum_j \|\Lambda(x_j y_j^\dagger)\|^2 - \mathrm{Re}\langle \Lambda(x_1 y_1^\dagger), \Lambda(x_2 y_2^\dagger)\rangle + |t|^2$ and expanding $\|\Lambda(\Delta)\|^2$ with (A2) yields

$$Q_{L|R}(C) = \frac{(n_1 - n_2)^2}{2} + |\omega|^2 + \frac{s_X + s_Y + s_1 + s_2 + \|\Lambda(\Delta)\|^2}{2} \geq 0. \quad \text{(A3)}$$

The identity holds for unequal dimensions and rank drops; equality forces $n_1 = n_2$, $\omega = 0$, $s_X = s_Y = s_1 = s_2 = 0$ and $\Lambda(\Delta) = 0$.

Appendix B: An endpoint bound in every dimension

The uniform certificate of Sec. VII has one hand input, the following quantitative bound on the endpoint form, valid in every dimension. We write $N = \|C\|^2$, $T = |\operatorname{Tr} C|^2$, $B_j = \sum_{|S|=j} \|\operatorname{Tr}_S C\|^2$, so that $q = N - \frac{1}{2}B_1 + \frac{1}{4}B_2 - \frac{1}{8}T$, and $g = (s_1 - s_2)^2$ for the singular values $s_1 \geq s_2 \geq 0$ of C (a rank-one C has $s_2 = 0$).

Theorem B.1. *For every C of rank at most two on $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$ of arbitrary finite dimensions,*

$$q(C) \;\geq\; -\tfrac{1}{16}N + \tfrac{1}{32}T + \tfrac{1}{8}g, \tag{B1}$$

i.e. $\frac{17}{8}N - B_1 + \frac{1}{2}B_2 - \frac{5}{16}T \geq \frac{1}{4}g \geq 0$.

The proof combines four elementary bounds. Throughout, $\langle A, B\rangle = \operatorname{Tr} A^\dagger B$ and

$$q(A,B) = \sum_S (-\tfrac{1}{2})^{|S|} \langle \operatorname{Tr}_S A, \operatorname{Tr}_S B\rangle$$

is the polarisation of q.

Step 1: positive matrices. Let $P \succeq 0$ have rank at most two and eigenvalues $\lambda \geq \mu \geq 0$. Then

$$q(P) \geq \tfrac{1}{8}(\lambda - \mu)^2 = \tfrac{1}{8}\big(2\|P\|^2 - (\operatorname{Tr} P)^2\big). \tag{B2}$$

By homogeneity take $\operatorname{Tr} P = 1$ and write $P = W\tau W^\dagger$ with $W^\dagger W = I_2$, $\tau = \operatorname{diag}(\lambda, \mu)$. With the Pauli matrices $\varsigma_1, \varsigma_2, \varsigma_3$ put $K_\nu = \sqrt{\tau}\,\varsigma_\nu \sqrt{\tau}$ and $M_{i\nu} = \operatorname{Tr}_{\bar{i}}(W K_\nu W^\dagger)$, an operator on $\mathcal{H}_i$. In the unit vector $\psi = \sum_a \sqrt{\tau_a}\,|a\rangle \otimes W|a\rangle \in \mathbb{C}^2 \otimes \mathcal{H}$, the Hermitian operator $L_{i\nu} = \varsigma_\nu^{\mathsf{T}} \otimes \operatorname{sgn}(M_{i\nu})$ has expectation $\operatorname{Tr}[\operatorname{sgn}(M_{i\nu}) M_{i\nu}] = \|M_{i\nu}\|_1$. For fixed $\nu - i$ modulo 3 the three operators $L_{i\nu}$ anticommute (the Pauli parts do, the sign parts act on different sites) and satisfy $L_{i\nu}^2 \leq I$, so $(\sum_i t_i L_{i\nu_i})^2 \leq \sum_i t_i^2$ for real t, whence $\sum_i \langle \psi, L_{i\nu_i}\psi\rangle^2 \leq 1$. Summing the three classes, $\sum_{i,\nu} \|M_{i\nu}\|_1^2 \leq 3$. For Hermitian M, $2\|M\|^2 \leq \|M\|_1^2 + (\operatorname{Tr} M)^2$, and $\operatorname{Tr} M_{i\nu} = \operatorname{Tr} K_\nu = (\lambda - \mu)\delta_{\nu 3}$; hence $\sum_{i,\nu} \|M_{i\nu}\|^2 \leq \frac{3}{2}\big(1 + (\lambda - \mu)^2\big) = 3\|P\|^2$. Finally, with $X_a = \sqrt{\tau_a} W|a\rangle$ and $\varsigma_0 = I$, Pauli completeness $\sum_{\nu=0}^3 (\varsigma_\nu)_{ab}(\varsigma_\nu)_{dc} = 2\delta_{ac}\delta_{bd}$ and the swap identity of Lemma 6.2 give $\sum_{\nu=0}^3 \|M_{i\nu}\|^2 = 2\sum_{a,b} \|\operatorname{Tr}_{\bar{i}}(X_a X_b^\dagger)\|^2 = 2\operatorname{Tr}[(P \otimes P) F_{\bar{i}}] = 2\|\operatorname{Tr}_i P\|^2$, i.e.

$$\textstyle\sum_{\nu=1}^3 \|M_{i\nu}\|^2 = 2\|\operatorname{Tr}_i P\|^2 - \|\operatorname{Tr}_{\bar{i}} P\|^2. \tag{B3}$$

Substituting (B3) into $q(P) = \|P\|^2 - \frac{1}{2}\sum_i \|\operatorname{Tr}_i P\|^2 + \frac{1}{4}\sum_i \|\operatorname{Tr}_{\bar{i}} P\|^2 - \frac{1}{8}$ gives $q(P) = \|P\|^2 - \frac{1}{4}\sum_{i,\nu} \|M_{i\nu}\|^2 - \frac{1}{8} \geq \frac{1}{8}(2\|P\|^2 - 1)$, which is (B2).

Step 2: four positive secants. Let $C = USV^\dagger$ with $U^\dagger U = V^\dagger V = I_2$, $S = \operatorname{diag}(s_1, s_2)$, and put $P = USU^\dagger$, $Q = VSV^\dagger$, $s = s_1 + s_2$, $f = \operatorname{Tr}(PQ)$, $G = 2f - T$; note $g = 2N - s^2$. For $z \in \{1, -1, i, -i\}$ the matrix $P_z = (U + zV)S(U + zV)^\dagger$ is positive of rank at most two, so $e(P_z) = q(P_z) - \frac{1}{8}\operatorname{gap}(P_z)^2 \geq 0$ by (B2), where gap is the difference of the two eigenvalues. Let further

$$\Delta = \sum_{i,\nu=1}^3 \big\|\operatorname{Tr}_{\bar{i}}\big[U\sqrt{S}\varsigma_\nu\sqrt{S}U^\dagger - V\sqrt{S}\varsigma_\nu\sqrt{S}V^\dagger\big]\big\|^2 \geq 0,$$

$$\mathcal{S} = \operatorname{Tr}\big[\tfrac{1}{2}(I + F_\ell)\, L^\dagger L\big] \geq 0, \qquad L = \Pi\,(U\sqrt{S} \otimes V\sqrt{S}),$$

where $\Pi = \prod_i \frac{1}{2}(I - F_i)$ on $\mathcal{H} \otimes \mathcal{H}$ and F_ℓ exchanges the two logical qubits (the column spaces of $U\sqrt{S}$ and $V\sqrt{S}$). Then

$$24\,q(C) - 4g + G = \sum_z e(P_z) + \Delta + 64\,\mathcal{S}, \tag{B4}$$

so that

$$q(C) \geq \tfrac{1}{24}(4g - G). \tag{B5}$$

Identity (B4) is a finite expansion; we list the five relations that produce it. Write $X_a = \sqrt{s_a}\,U|a\rangle$ and $Y_a = \sqrt{s_a}\,V|a\rangle$, so that $C = \sum_a X_a Y_a^\dagger$, $P = \sum_a X_a X_a^\dagger$, $Q = \sum_a Y_a Y_a^\dagger$, and put $E = q(P) + q(Q)$, $h = q(P, Q)$ and $c_j = \sum_{|J|=j} \operatorname{Tr}[(\operatorname{Tr}_{J^c} P)(\operatorname{Tr}_{J^c} Q)]$. Two contraction identities are used throughout: the polarised form $\langle \operatorname{Tr}_S(xy^\dagger), \operatorname{Tr}_S(x'y'^\dagger)\rangle = \operatorname{Tr}[F_S(x'x^\dagger \otimes yy'^\dagger)]$ of the rank-one identity of Sec. IV, and $\operatorname{Tr}[F_S(A \otimes B)] = \operatorname{Tr}[(\operatorname{Tr}_{S^c} A)(\operatorname{Tr}_{S^c} B)]$; together they give $\sum_{a,b}\langle \operatorname{Tr}_S(X_a X_b^\dagger), \operatorname{Tr}_S(Y_a Y_b^\dagger)\rangle = \operatorname{Tr}[F_S(C^\dagger \otimes C)] = \|\operatorname{Tr}_{S^c} C\|^2$. (1) $8h = 8f - 4c_2 + 2c_1 - s^2$ is the definition of h. (2) The identity $q(P) = \|P\|^2 - \frac{1}{4}\sum_{i,\nu}\|M_{i\nu}\|^2 - \frac{1}{8}(\operatorname{Tr} P)^2$ of Step 1 is homogeneous, so it holds for P with $M_{i\nu}^P = \operatorname{Tr}_{\bar{i}}(U\sqrt{S}\varsigma_\nu\sqrt{S}U^\dagger)$ and for Q with $M_{i\nu}^Q$ built from V; the polarisation of (B3) and the contraction identity give $\sum_{\nu=1}^3 \langle M_{i\nu}^P, M_{i\nu}^Q\rangle = 2\|\operatorname{Tr}_i C\|^2 - \operatorname{Tr}[(\operatorname{Tr}_{\bar{i}} P)(\operatorname{Tr}_{\bar{i}} Q)]$. Since $\|P\|^2 = \|Q\|^2 = N$ and $\Delta = \sum_{i,\nu}\|M_{i\nu}^P - M_{i\nu}^Q\|^2$, this yields $4E + \Delta = 8N - s^2 - 4B_1 + 2c_1$. (3) $\Pi = \frac{1}{8}\sum_S (-1)^{|S|} F_S$, so $\operatorname{Tr}(L^\dagger L) = \operatorname{Tr}[\Pi(P \otimes Q)] = \frac{1}{8}(s^2 - c_1 + c_2 - f)$, and $\operatorname{Tr}(F_\ell L^\dagger L) = \sum_{a,b}\langle X_b \otimes Y_a, \Pi(X_a \otimes Y_b)\rangle = \frac{1}{8}\sum_S (-1)^{|S|}\|\operatorname{Tr}_S C\|^2$; hence $64\mathcal{S} = 4(s^2 + N - c_1 + c_2 - f - B_1 + B_2 - T)$. (4) $P_z = P + Q + zC^\dagger + \bar{z}C$, and averaging over the four phases kills the terms of odd degree and those in $z^2, \bar{z}^2$: $\sum_z q(P_z) = 4q(P+Q) + 4q(C) + 4q(C^\dagger) = 4E + 8h + 8q(C)$. (5) The nonzero eigenvalues of P_z are those of $2S + zZ + \bar{z}Z^\dagger$ with $Z = \sqrt{S}\,U^\dagger V\sqrt{S}$, $|\operatorname{Tr} Z|^2 = T$ and $\|Z\|^2 = f$; with $\operatorname{gap}(M)^2 = 2\operatorname{Tr} M^2 - (\operatorname{Tr} M)^2$ and the same averaging, $\sum_z \operatorname{gap}(P_z)^2 = 16g + 8G$. Adding (1)–(5) gives (B4).

Step 3: Hermitian parts. Let Y be Hermitian with at most two eigenvalues of each sign, $Y = A - B$ with $A, B \succeq 0$ of rank at most two and $AB = 0$. Positivity of $\prod_i (I - F_i)$ on $A \otimes B$ gives $\sum_E (-1)^{|E|} \operatorname{Tr}[(\operatorname{Tr}_{E^c} A)(\operatorname{Tr}_{E^c} B)] \geq 0$, i.e. $b \leq u + a$ with $u = \operatorname{Tr} A \operatorname{Tr} B$, $a = \sum_i \operatorname{Tr}[(\operatorname{Tr}_i A)(\operatorname{Tr}_i B)]$, $b =$

$\sum_i \mathrm{Tr}[(\mathrm{Tr}_{\bar{i}} A)(\mathrm{Tr}_{\bar{i}} B)]$; hence $q(A,B) = -\frac{1}{2}a + \frac{1}{4}b - \frac{1}{8}u \le \frac{1}{8}u$. With (B2) for A and B,

$$q(Y) = q(A) + q(B) - 2q(A,B) \ge \tfrac{1}{8}\big(2\|Y\|^2 - \|Y\|_1^2\big). \tag{B6}$$

If $\operatorname{rank} C \le 2$, the quadratic forms of $\operatorname{Re} C = \frac{1}{2}(C + C^\dagger)$ and $\operatorname{Im} C = \frac{1}{2i}(C - C^\dagger)$ vanish on $\ker C$, a subspace of codimension at most two, so each has at most two eigenvalues of either sign; and $q(C) = q(\operatorname{Re} C) + q(\operatorname{Im} C)$ because $q(A,B)$ is real for Hermitian A, B. Therefore

$$q(C) \ge \tfrac{1}{8}\big(2N - \|\operatorname{Re} C\|_1^2 - \|\operatorname{Im} C\|_1^2\big). \tag{B7}$$

Step 4: a Bessel inequality. Let $J = \operatorname{sgn}(\operatorname{Re} C)$ (a Hermitian unitary, with any signs on $\ker \operatorname{Re} C$), $E_0, \dots, E_3$ a Hermitian orthonormal basis of $M_2(\mathbb{C})$ and $Z = \sqrt{S}\,U^\dagger V \sqrt{S}$, so that $f = \|Z\|^2$. For the real inner product $\operatorname{Re}\mathrm{Tr}(A^\dagger B)$ the five matrices $JU/\sqrt{2}$, $iUE_0, \dots, iUE_3$ are orthonormal, since $U^\dagger JU$ and the E_μ are Hermitian. Bessel's inequality for VS gives $\frac{1}{2}\|\operatorname{Re} C\|_1^2 + \|\operatorname{Im}(U^\dagger V S)\|^2 \le N$, using $\|\operatorname{Re} C\|_1 = \operatorname{Re}\mathrm{Tr}(JUSV^\dagger)$. Exchanging U and V, averaging, and using the entrywise identity $\|\operatorname{Im}(MS)\|^2 + \|\operatorname{Im}(SM)\|^2 - 2\|\operatorname{Im}(\sqrt{S}M\sqrt{S})\|^2 = \frac{1}{2}\sum_{jk}(S_{jj} - S_{kk})^2 |M_{jk}|^2 \ge 0$ with $M = U^\dagger V$ gives $\|\operatorname{Re} C\|_1^2 \le 2N - 2\|\operatorname{Im} Z\|^2$. The same argument for iC gives $\|\operatorname{Im} C\|_1^2 \le 2N - 2\|\operatorname{Re} Z\|^2$, and adding,

$$\|\operatorname{Re} C\|_1^2 + \|\operatorname{Im} C\|_1^2 \le 4N - 2f. \tag{B8}$$

Conclusion. By (B7) and (B8), $q(C) \ge \frac{1}{8}(G + T - 2N)$; one quarter of this bound plus three quarters of (B5) is (B1). The expansions of Step 2 and every inequality above are also checked numerically on random instances in the package [20].

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